■ FarmWise360 AI

Complete System Architecture

Master Model · Slave Model · NLP · NLU · Neural Networks

Project	FarmWise360 AI — Smart Agricultural Advisory
Team	Thanya Ravi & Vachan Gowda V · TIMES POINT
Version	v2.0 — Master-Slave Architecture
Date	October 2026
Stack	NLP + NLU + Neural Networks (< 1M params offline)
Philosophy	Vast Knowledge · Tiny Size · Zero Hallucination · Always Safe

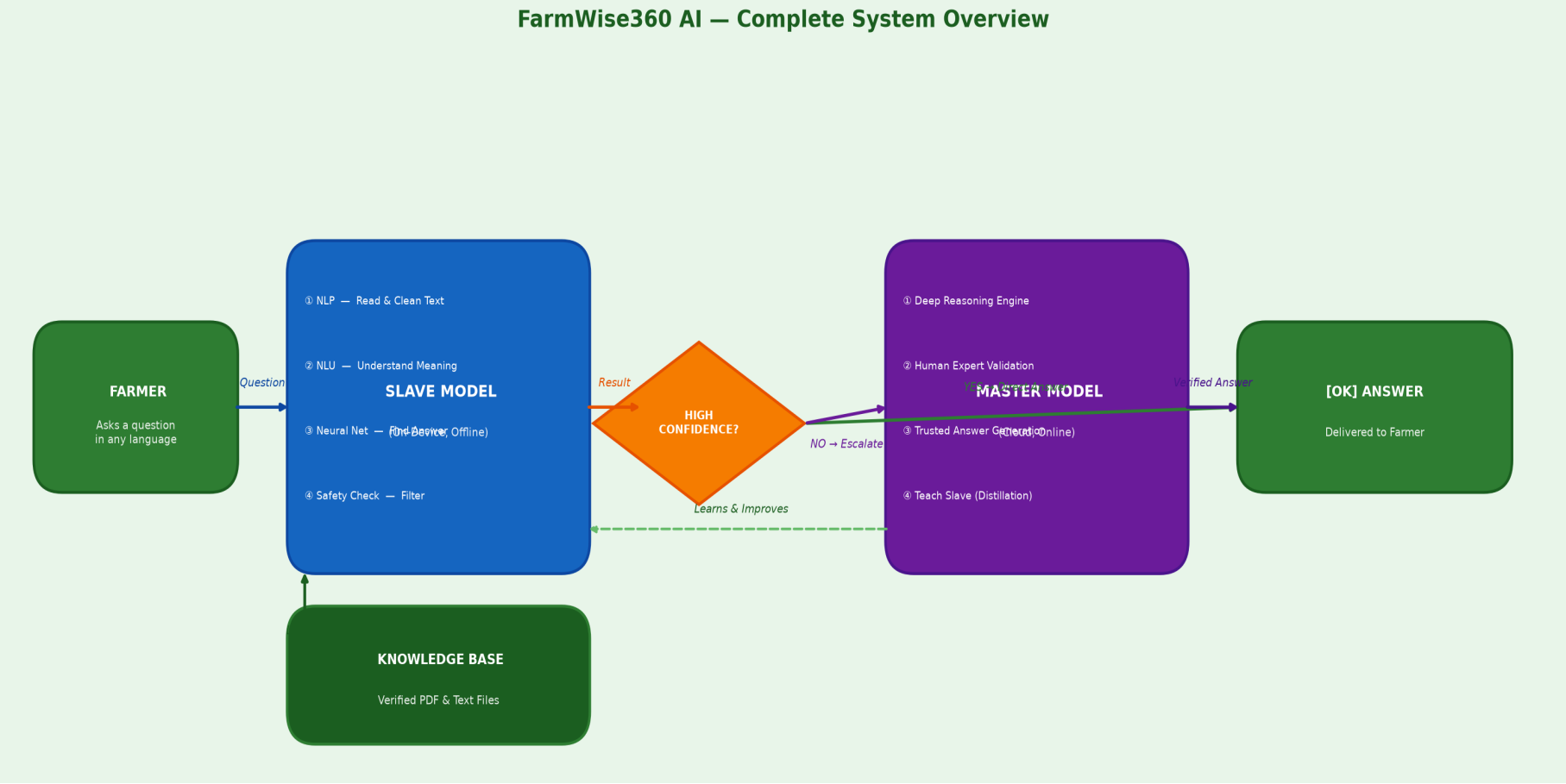
This document explains — in plain language — how FarmWise360 AI works from the moment a farmer asks a question to the moment a safe, verified answer reaches their phone. No jargon. No shortcuts. Just clear diagrams and honest explanations.

[LIST] Table of Contents

Page	Section	What it covers
3	System Overview	The big picture — how everything connects
4	Slave Model Pipeline	Step-by-step: how offline device answers
5	Master Model Pipeline	Deep reasoning + human validation in cloud
6	Data Processing Pipeline	How PDFs & TXT files become instant answers
7	Safety & Legal Framework	4 gates every answer must pass
8	Key Design Decisions	Why we built it this way

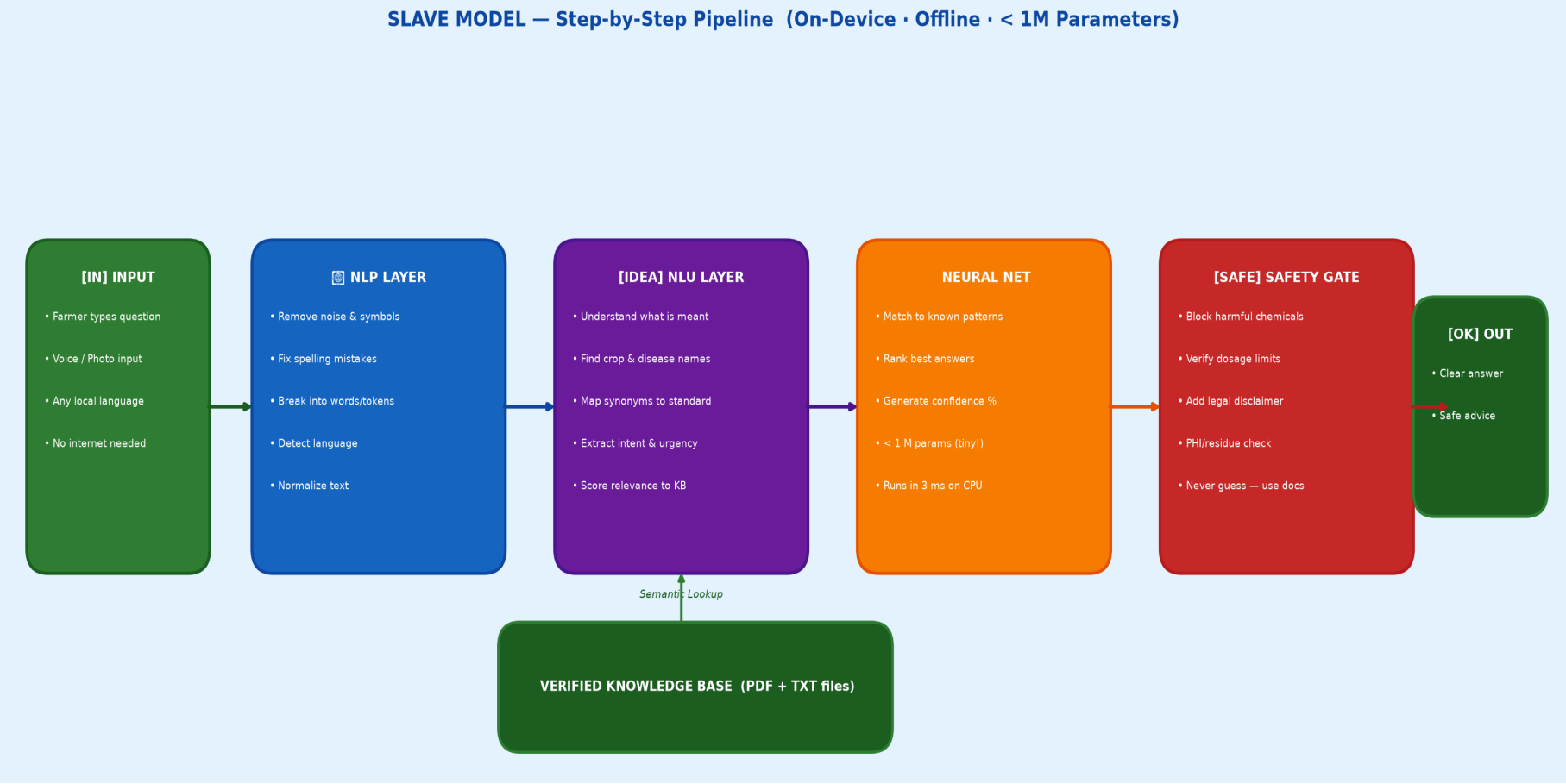
■ Section 1 — System Overview: The Big Picture

Think of FarmWise360 like a two-doctor clinic. The **Slave Model** is the **local doctor** who sits in the village — always available, no internet needed, answers common questions in seconds. When the question is too complex, it calls the **Master Model** — the **specialist doctor** in the city who does deep research and also has a human expert double-check the answer before it's sent. The specialist then teaches the local doctor so it keeps getting smarter.



Section 2 — Slave Model: Step-by-Step Pipeline

The Slave Model runs **entirely on the farmer's device — no internet required**. It is tiny (under 1 million parameters) yet powerful enough to handle 332 questions per second with a 3 ms response time. Here is every step:

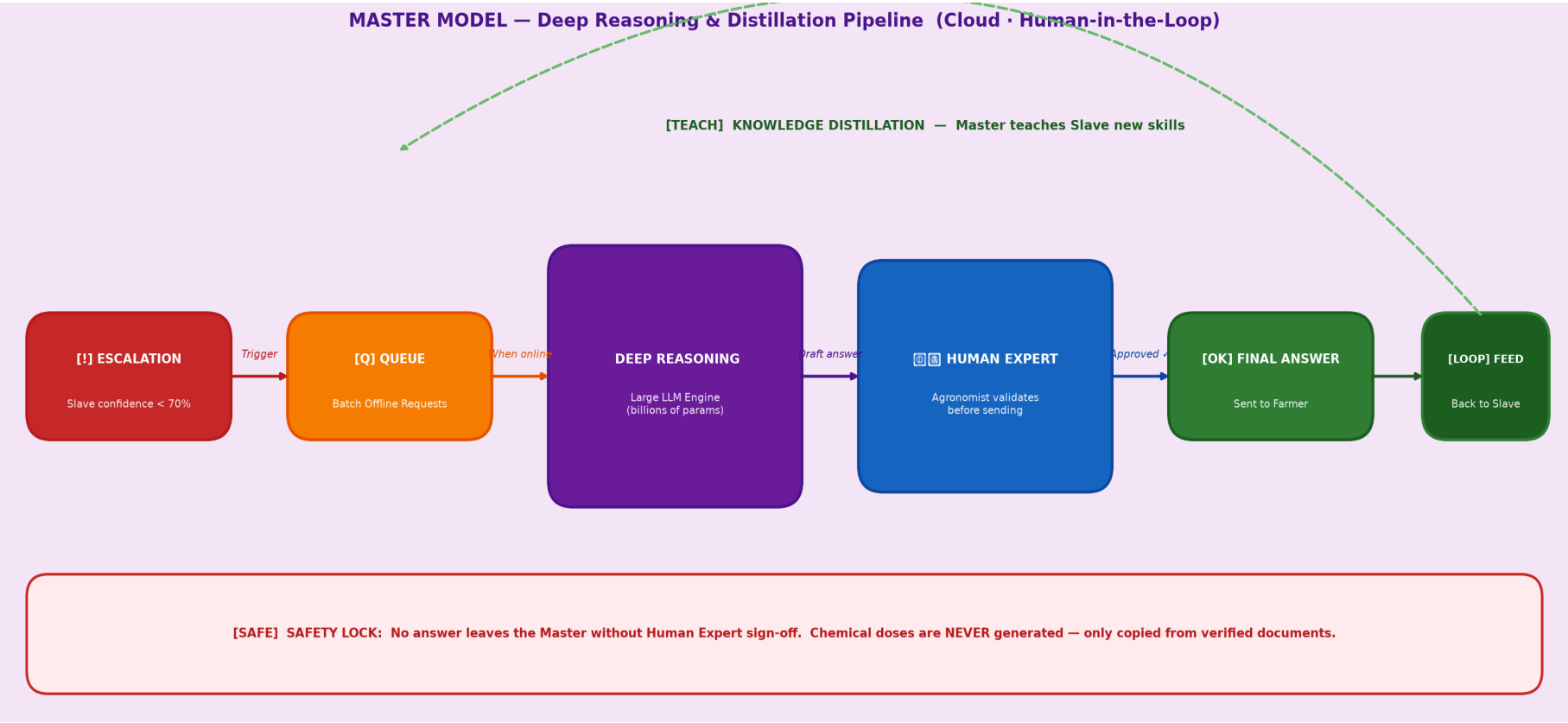


Step	Layer	What happens in plain language
①	NLP — Natural Language Processing	Reads the farmer's raw text, fixes typos, removes punctuation noise, identifies the language, and breaks it into clean word pieces (tokens).

②	NLU — Natural Language Understanding	Goes deeper than words — figures out WHAT the farmer means. Maps colloquial names ('yellowing leaf') to scientific terms, extracts the cr
③	Neural Network	Matches the understood question against learned patterns from the Knowledge Base. Produces a confidence score (0-100 %). Under 1M pa
④	Safety Gate	Every answer passes through a filter that blocks banned chemicals, verifies dosages against verified documents, and adds a legal disclaime

Section 3 — Master Model: Deep Reasoning in the Cloud

When the Slave Model is not sure (confidence below 70 %), it queues the question for the **Master Model in the cloud**. The Master uses a large, powerful AI engine and — critically — always has a **human agricultural expert validate the answer** before it reaches the farmer. No answer from the Master is automatic.

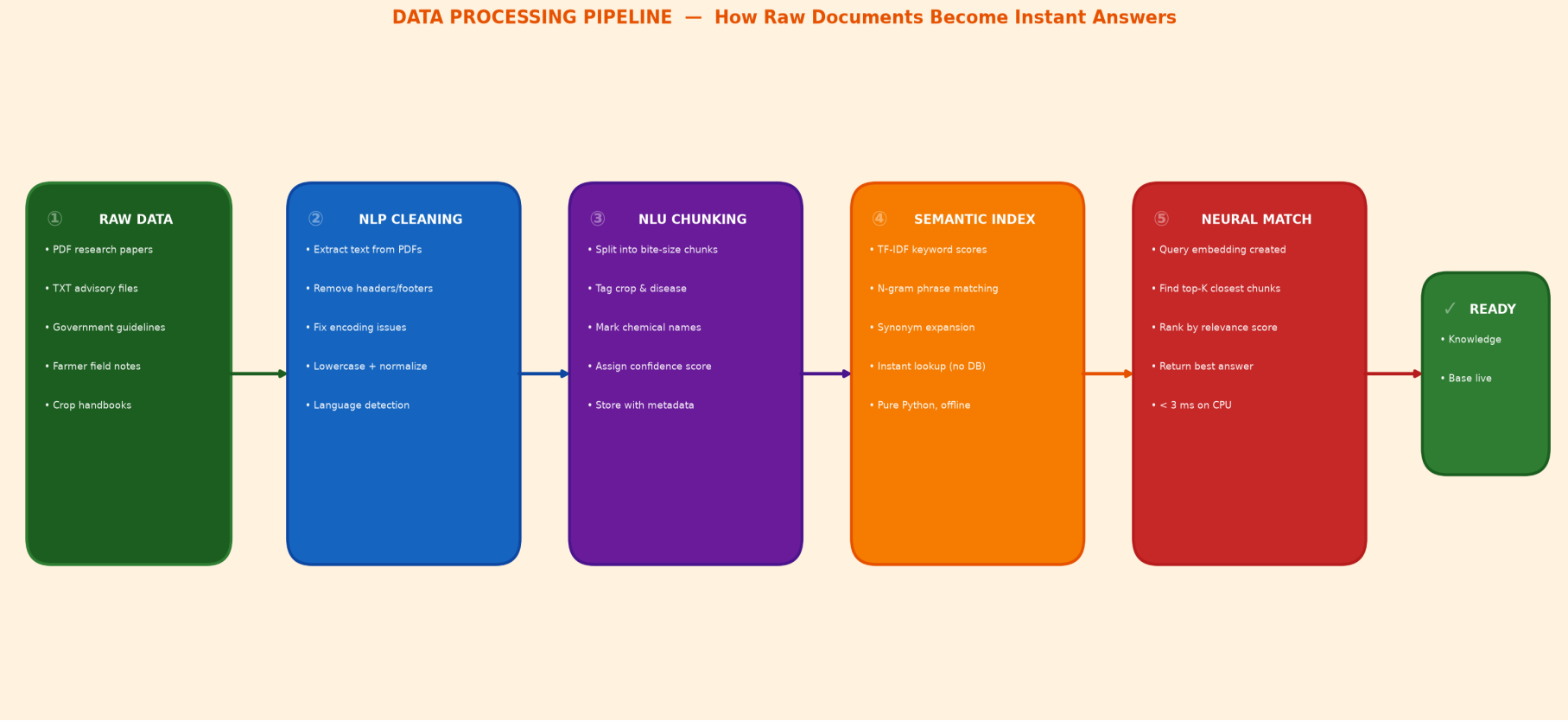


Step	What happens
Escalation Trigger	Slave confidence < 70% → question is queued, farmer gets 'checking with expert' message
Offline Queue	Questions are stored locally and synced when internet is available — no data is lost

Deep Reasoning	Large language model (billions of params) analyses the question with full agricultural context
Human Expert Review	A certified agronomist reads the AI draft and approves, edits, or rejects it
Answer Delivery	Only after expert sign-off does the verified answer reach the farmer
Knowledge Distillation	The Master's new knowledge is compressed and taught back to the Slave so it improves over time

Section 4 — Data Processing: PDF → Instant Answers

FarmWise360 is built on **real documents — PDFs, government guidelines, research papers, and advisory texts** — not on noisy, unverified CSV datasets. Here is how raw documents are turned into the fast, accurate Knowledge Base that powers every answer:

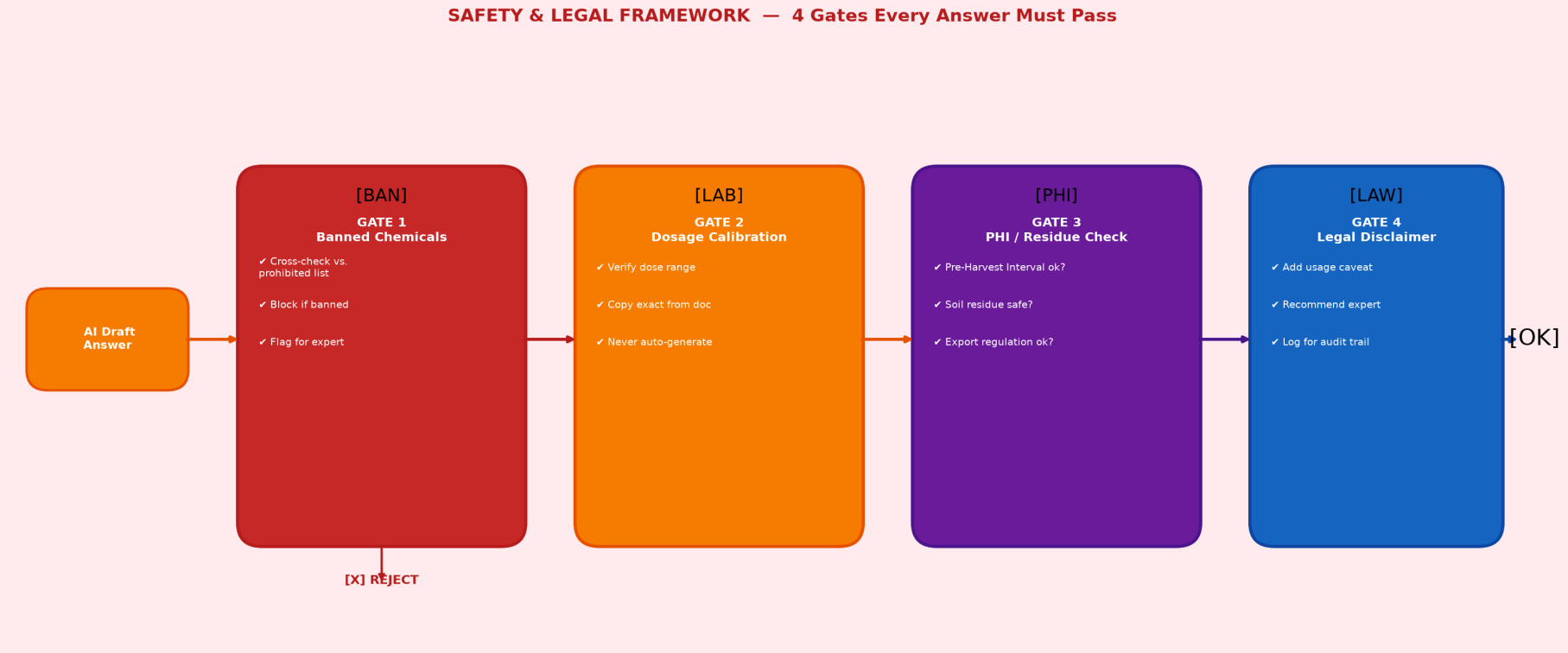


Stage	Tool / Method	Plain-language explanation
① Collect	PDF + TXT files	Trusted sources only: ICAR, TNAU, state agriculture departments, peer-reviewed papers
② NLP Clean	pypdf + regex + langdetect	Text is extracted from PDFs, headers/footers removed, language detected, and text normalised

③ NLU Chunk	Sentence splitter + crop tagger	Long documents are cut into small, meaningful paragraphs each tagged with crop, disease, and chemical metadata
④ Index	TF-IDF + N-gram + synonyms	Each chunk gets a relevance score. Synonyms are expanded so 'yellowing' also matches 'chlorosis'. No external database needed
⑤ Match	Neural embedding lookup	When a question arrives, the Neural Net finds the top-matching chunks in under 3 ms and assembles the answer

[SAFE] Section 5 — Safety & Legal Framework: 4 Non-Negotiable Gates

Every single answer — whether from Slave or Master — must pass through **4 safety gates** before reaching the farmer. These gates are **non-negotiable and cannot be bypassed**. This protects farmers, protects the environment, and protects the company legally.



Gate	Name	What it does	What happens if it fails
1	Banned Chemical Filter	Every chemical name in the answer is checked against the national list of banned pesticides.	Answer is BLOCKED. Farmer is told to consult local authority.
2	Dosage Calibration	Dosages are NEVER generated by the AI. They are copied verbatim from approved documents.	Dosage is removed. Only qualitative advice is given.
3	PHI / Residue Check	Pre-Harvest Interval (PHI) and soil residue rules are verified so the farmer's produce does not fail export standards.	Additional warnings are added to the advice.
4	Legal Disclaimer	Every answer includes a standard disclaimer recommending the farmer consult a local agronomist.	Answer is not sent without disclaimer. All gates are logged for audit.

[IDEA] Section 6 — Key Design Decisions: Why We Built It This Way

■ Why NLP + NLU + Neural Networks?

NLP cleans the language, NLU understands the meaning, and the Neural Network finds the best answer. These three layers together give us accuracy without requiring a huge, expensive model.

■ Why PDF and TXT files instead of spreadsheet datasets?

Government guidelines and research papers in PDF/TXT format are already trusted, structured, and widely available in agriculture. They contain more reliable information than auto-generated CSV rows and are free to use.

■ Why is the Slave Model under 1 million parameters?

A small model fits on a basic Android phone, runs without internet, consumes very little battery, and still responds in 3 ms. Farmers in remote areas benefit the most from this tiny-but-smart design.

■ Why is there always a human in the loop for the Master?

Agriculture advice can affect a farmer's livelihood and health. No AI — however advanced — should send chemical recommendations without a certified expert confirming them. The human gate is mandatory, always.

■ Why does the Master teach the Slave (Knowledge Distillation)?

Every time the Master solves a hard case, that knowledge is compressed and added to the Slave. This means the Slave gets smarter with every deployment cycle without becoming bigger or slower — vast knowledge, tiny size.

■ How does zero-hallucination work in practice?

All dosages and chemical names are stored verbatim in the verified Knowledge Base. The model is NOT allowed to generate these — it can only look them up. This is enforced at the Safety Gate level, not just as a rule.

Performance at a Glance

Metric	Slave Model	Master Model

Parameters	< 1 Million	Billions (cloud)
Response Time	3 ms (CPU)	5–30 seconds (cloud + human review)
Internet Required?	No — fully offline	Yes — escalation path only
Human in the Loop?	No (direct answer)	Yes — mandatory
Use Case	Common, known questions	Complex, rare, or high-risk queries